FENCES DO NOT MAXIMISE ORBITS OF LINEAR EXTENSIONS

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ABSTRACT. For a finite poset P let $f(P) = e(P)/\#\mathrm{Aut}(P)$ be the number of orbits of the automorphism group on the linear extensions, and let $g(n)$ be the largest value of f over n-element posets. Stanley asked what can be said about $g(n)$ and whether the maximisers are disjoint unions of fences. By exhaustive computation we find $g(n)$ and all maximisers for $n \le 11$: fence unions are maximisers exactly for $n \le 7$ and $n = 9$, and the first failure is an eight-element poset with $f = 2384$ against 2100 for the best fence union. We also prove that for every $n \ge 58$ no maximiser is a union of fences. The proof combines a construction posted by Graham with an upper bound for fence unions: a poset with t disjoint comparable pairs has at most $n!/2^t$ linear extensions.

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1. INTRODUCTION

A linear extension of a finite poset P is a way of listing its elements so that every relation $x < y$ puts x before y. The automorphism group of P permutes these listings, and only the identity fixes one, so every orbit has exactly $\#\mathrm{Aut}(P)$ elements. The number of orbits is therefore

$$f(P) = \frac{e(P)}{\#\mathrm{Aut}(P)},$$

where $e(P)$ is the number of linear extensions. Large $e(P)$ wants few relations, but few relations bring symmetry: the antichain has $e = n!$ and $f = 1$. Stanley [3] asked how large f can be. Writing $g(n)$ for the maximum over n-element posets, what can be said about $g(n)$ and the posets attaining it, and in particular, are they disjoint unions of fences?

A *fence* Z_m is the poset $x_1 < x_2 > x_3 < x_4 > \cdots$ on m elements; for odd $m \ge 3$ its dual is a different poset, and we count both as fences. A linear extension of Z_m gives positions $L(x_1) < L(x_2) > L(x_3) < \cdots$, which is an alternating permutation, so $e(Z_m) = E_m$, the Euler zigzag number [4]. Fences are a natural guess: they have many linear extensions for the number of relations, and little symmetry.

A first case. For $n = 5$ the maximiser is $Z_4 + \bullet$, a four-element fence and a separate point: $e = 5 \cdot E_4 = 25$ and the automorphism group is trivial, so $f = 25$. At $n = 8$ this pattern breaks. The best union of fences is $Z_2 + Z_4 + 2\bullet$, with $e = 4200$ and two automorphisms (swap the points), so $f = 2100$. The poset $Q + \bullet$ of Figure 1, a six-element fence with one extra element below its middle peak, plus a point, has $e = 8 \cdot 298 = 2384$ and no automorphisms.

Theorem 1. *For $n \le 11$ the values of $g(n)$ are*

$$1,\ 1,\ 3,\ 6,\ 25,\ 75,\ 525,\ 2384,\ 15372,\ 107280,\ 798435.$$

A disjoint union of fences attains $g(n)$ exactly when $n \le 7$ or $n = 9$. The maximisers for $n = 8, 10, 11$ are $Q + \bullet$, $Q + Z_2 + \bullet$ and $R + Z_2 + \bullet$ (with the duals of the first two), where R is Q with one more element above c.

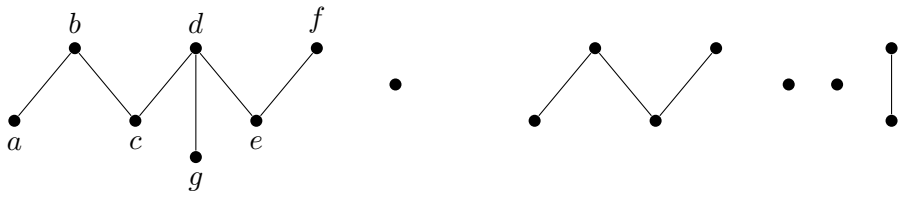
Theorem 1 is a finite computation (Section 5). The second result settles the fence question for all large n.

Theorem 2. *For every $n \ge 58$, no poset attaining $g(n)$ is a disjoint union of fences.*

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$Q + \bullet$: $f = 2384$ $\qquad$ $Z_4 + Z_2 + 2\bullet$: $f = 2100$

FIGURE 1. Eight elements. Left: the maximiser $Q + \bullet$ (and its dual), not a union of fences, since d is comparable with three elements. Right: the best union of fences. The extra element g adds relations but removes the reflection of the fence, and that trade wins.

The lower bound comes from Graham's answer [1] to Stanley's question: a few maximal elements over many minimal ones, wired so that the only symmetry permutes the top. Graham estimated that this beats tree-like posets for large n. The new ingredient is a bound that makes the comparison with fences rigorous and explicit (Proposition 2); for $58 \le n \le 180$ it is completed by an exact computation of the best fence union. Between 12 and 57 the question stays open.

2. An upper bound for unions of fences

The bound rests on one counting fact.

Lemma 1. *If P has n elements and contains t pairwise disjoint pairs $a_i < b_i$, then $e(P) \le n!/2^t$.*

Proof. Order the elements uniformly at random. For disjoint pairs the events "a_i comes before b_i" are independent, each of probability $\frac{1}{2}$, so all t hold for exactly $n!/2^t$ orderings. Every linear extension is one of them. □

In words: each disjoint relation at least halves the number of admissible orderings. A fence Z_m contains the $\lfloor m/2 \rfloor$ disjoint pairs $x_1 < x_2$, $x_3 < x_4$, …, and $\lfloor m/2 \rfloor \ge m/3$ when $m \ge 2$.

Proposition 2. *Let P be a disjoint union of fences on n elements, k of which are isolated points. Then*

$$f(P) \le \frac{n!}{2^{(n-k)/3}\, k!}.$$

Proof. The non-singleton fences cover $n - k$ elements and contain at least $(n - k)/3$ disjoint comparable pairs, so $e(P) \le n!/2^{(n-k)/3}$ by Lemma 1. Any permutation of the k isolated points is an automorphism, so $\#\mathrm{Aut}(P) \ge k!$. □

The bound reads: a union of fences loses a factor $2^{1/3}$ per element in a fence, and a factor $k!$ for its isolated points. Whatever k is, the combined loss is at least $2^{(n-1)/3}$, exponential in n (Section 4). The construction below loses only a factor $n^{\log_2 n + 1}$, which grows more slowly than any exponential.

For the exact comparison in the middle range we also need the exact value. The components of a fence union are fences Z_m and, for odd $m \ge 3$, their duals Z_m^*. One has $\#\mathrm{Aut}(Z_m) = 2$ for odd $m \ge 3$ (the reflection $x_i \mapsto x_{m+1-i}$) and 1 otherwise, since an automorphism of a path is the identity or the reflection, and for even m the reflection reverses the relations. For

a union with c copies of a component C of size m, the standard shuffle and wreath-product counts give

$$f(P) = n! \prod_{C} \frac{1}{c_C!} \Big(\frac{E_m}{m! \, \#\mathrm{Aut}(C)} \Big)^{c_C}. \tag{1}$$

Each component contributes its share $E_m/m!$ of orderings, divided by its own symmetry, and c identical components contribute a further $c!$ for permuting them. Maximising (1) over multisets of components is a knapsack over n, computed exactly in rational arithmetic.

3. Graham's construction

Lemma 3 (Graham [1])**.** *Let $j \geq 1$ and $1 \leq m \leq 2^j - 1$. There is a poset on $n = j + m$ elements with $f \geq m!$.*

Proof. Take j top elements A and m bottom elements b_S, one for each set S in a family of m distinct nonempty subsets of A that contains A itself, with $b_S < a$ exactly when $a \in S$. Listing all bottoms first, in any order, and then all tops, in any order, gives $m!\, j!$ linear extensions. Each bottom lies below some top and each top lies above b_A, so bottoms are exactly the minimal elements that are not maximal, and every automorphism maps tops to tops. An automorphism is determined by its action on A, because b_S is the only bottom whose up-set is S. So $\#\mathrm{Aut} \leq j!$ and $f \geq m!$. □

What pins down the tops is that A itself is one of the sets: every top then lies above b_A. The empty set could be allowed too (an isolated bottom $b_\emptyset$ is still minimal and still the only bottom with its up-set, and the Lean proof does without the nonempty condition), but nothing is gained for Theorem 2. Figure 2 shows a small case.

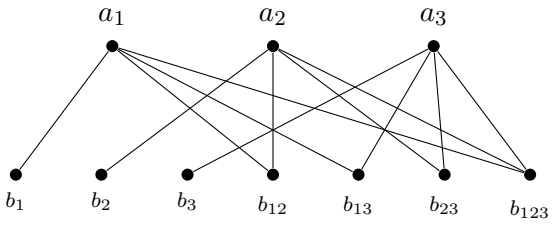


Figure 2. Graham's construction with $j = 3$ tops and all $m = 7$ nonempty subsets as bottoms ($n = 10$). Any order of the seven bottoms followed by any order of the three tops is a linear extension, so $e \geq 7!\,3!$, while the only symmetries permute the tops, so $\#\mathrm{Aut} \leq 3!$ and $f \geq 7! = 5040$. For larger n the bottoms outnumber the tops exponentially, and f is within a factor $n^{\log_2 n + 1}$ of $n!$.

For given n let $J(n)$ be the least j with $2^j + j - 1 \geq n$; then $g(n) \geq (n - J(n))!$. Since $J(n) \leq \log_2 n + 1$, this is $n!$ divided by a factor at most $n^{\log_2 n + 1}$.

4. Proof of Theorem 2

Fix n and a union of fences P with k isolated points; put $j = J(n)$. It is enough to show $f(P) < (n - j)!$.

Large n. By Proposition 2 it suffices that $k!\, 2^{(n-k)/3} > n!/(n-j)!$ for every k. The left side is smallest at $k \in \{0, 1\}$, where it is at least $2^{(n-1)/3}$, because each further step multiplies it by $(k+1)\, 2^{-1/3} \geq 1$. So it suffices that

$$\Big(\frac{n!}{(n-j)!} \Big)^3 < 2^{\,n-1}. \tag{2}$$

The left side is at most $n^{3j} \le n^{3(\log_2 n+1)}$, and $(n-1)/3 > (\log_2 n+1)\log_2 n$ holds at $n = 200$ and its difference grows with n, so (2) holds for all $n \ge 200$. Checking (2) exactly with integers shows it holds for every $181 \le n \le 199$ as well (and fails at $n = 180$).

Middle range. For $58 \le n \le 180$ we compute the exact maximum of (1) over all fence unions on n points and compare it with $(n - J(n))!$; the construction wins at every n in this range. For $12 \le n \le 57$ it loses at every n, so this argument says nothing there. □

5. The computation behind Theorem 1

Every poset on at most 11 elements was generated by nauty's `genposetg` [2] (46,749,427 posets at $n = 11$; the counts for $n \le 11$ match OEIS A000112). For each, $e(P)$ was computed by dynamic programming over down-sets and $\#\mathrm{Aut}(P)$ by backtracking over the Hasse diagram. Integrality of $e(P)/\#\mathrm{Aut}(P)$ was asserted for every poset, as the free action requires. For $n \le 8$ the whole search was repeated independently in Sage, with the same maxima and maximisers. The best fence unions for $n \le 11$, from (1), agree with $g(n)$ exactly when $n \le 7$ or $n = 9$, and are 2100, 76860 and 704550 for $n = 8, 10, 11$.

Remark 1. The sequence $g(n)$ is not in the OEIS (searched 23 October 2026). Lemma 3 and the bound $f(P) \le e(P) \le n!$ give $n!/n^{\log_2 n+1} \le g(n) \le n!$. Whether fence unions ever maximise for $12 \le n \le 57$ is open; the computed maximisers suggest that a fence with pendant elements, as in Q and R, is the competitor to beat in that range.

Verification. The accompanying repository contains the search programs for $n \le 11$ with the Sage cross-check; an exact computation of the best fence union for every $n \le 200$ with two independent programs that share no code; brute-force checks of $e(Z_m) = E_m$ and of $\#\mathrm{Aut}(Z_m)$ for $m \le 8$; and an exact check of (2). In Lean 4 with Mathlib are formalised: the free action and $f = e/\#\mathrm{Aut}$; the posets Q, $Q{+}\bullet$, R and their counts; Lemma 1, Proposition 2, Lemma 3, and Theorem 2 for $n \ge 181$. The exhaustive search and the range $58 \le n \le 180$ are computations.

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